

September 23, 2024

Dear ICES Journal Of Marine Science Editorial Office,

We are pleased to submit our original article, “Identifying and prioritising key data gaps for modelling Antarctic krill recruitment” to the ICES Journal of Marine Science (JMS), Zooplankton Production Symposium special issue. In this study, we aimed to significantly improve the future development of krill models by providing a comprehensive overview of the data available for parameterising krill recruitment models. We identified model mechanisms with parameterisation information that is abundant, as well as those that would benefit most from additional experiments or measurements. To do so, we extracted >3500 data points from 254 studies which were selected by a systematic literature review. In addition to creating a “map of the data landscape” for developing krill recruitment models, we compiled an extensive data catalogue that can be readily utilized for future model development.

Antarctic krill is a key species in the Southern Ocean (SO), and krill populations are under increasing pressure due to a growing fishery, climate change and recovering baleen whale populations. Having robust models to project the potential responses of krill populations to these stressors is critical not only for fisheries management but also for our overall understanding of the SO ecosystem. Therefore, we believe that our study will be of broad interest to the SO modelling community and inspire similar efforts in other fields. Our submitted study aligns the scope of the ICES JMS to publish studies that contribute to understanding marine systems, address management and conservation issues, and benefit future interdisciplinary efforts. In addition, Early Career Researchers led this study’s design and execution, with senior scientist support, reflecting ICES JMS’s commitment to career development and diverse perspectives.

Finally, this work was presented and extensively discussed in session 14 at the Zooplankton Production Symposium in March 2024, and we believe that the ICES JMS ZPS special issue would be a perfect fit for our study. This work was constructed as a set of two sister papers identified as a high priority by the Antarctic krill modelling community. The second paper, “Untangling the Complexities of Larval Antarctic Krill Overwintering Success Under Climate Change”, is also planned for submission to the special issue.

We suggest the following three referees who possess relevant expertise in Antarctic krill modelling and zooplankton research but who have not been associated with the research or closely affiliated with any of the authors in the past five years:

- Nora-Charlotte Pauli, <nora-charlotte.pauli@awi.de>
- Kanchana Bandara, <kba@akvaplan.niva.no>
- Haiting Zhang, <haiting.zhang@utas.edu.au>

Thank you for your consideration.

Sincerely,

Alexis A. Bahl and Dominik Bahlburg